



ODONT 541.1

CARIOLOGY

Other Active Ingredients and Minimally Invasive Options Lecture #11

Course Director: Sahel Farhangi, DDS

Office Hours: By appointment
(Tuesday and Wednesday afternoons)

Mission: To advance the science and art of health through education, service, scholarship and social accountability



Other Active Ingredients

- Fluorides (previously covered)
 - CCP-ACP
 - Potassium nitrate
 - Strontium
 - Nanohydroxyapatite
 - SDS/SLS
 - Xylitol
 - Chlorhexidine
 - Chlorine Dioxide
-
- Resin infiltration (a technique, not an ingredient)

Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission

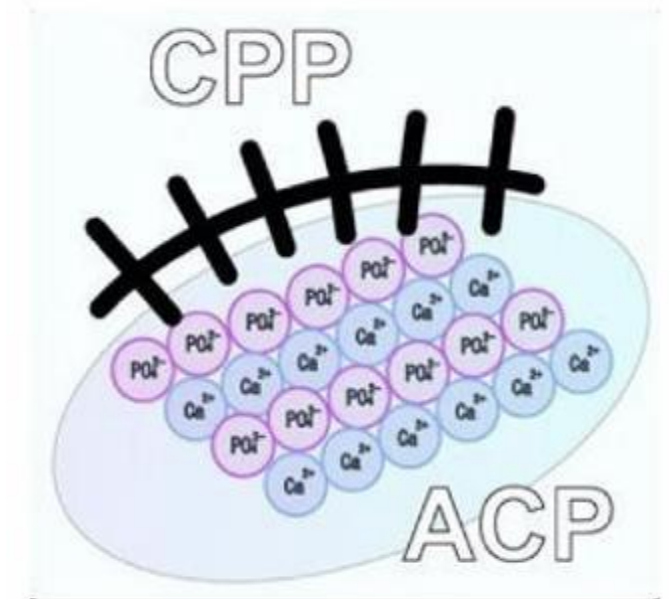
Calcium and Phosphate

- We've stated that the critical pH for hydroxyapatite is 5.5
 - But in reality, critical pH is not fixed.
 - **Critical pH is the point at which the demineralization rate is equal to the remineralization rate.**
 - Critical pH varies depending on the concentration of calcium and phosphate in the saliva and plaque fluid.
 - If there is excess calcium and phosphate available (fluids are supersaturated), demineralization is unlikely.
- Available calcium and phosphate (not just fluorida) are required for remineralization too.

CPP - ACP

(Casein Phosphopeptide – Amorphous Calcium Phosphate)

- Even in patient's with normal salivary flow, low levels of bioavailable calcium and phosphate can become limiting factors for remineralization.
- CPP (the black comb)
 - Derived from milk protein
 - Milk is minimally effective due to large proteins and minimal availability of free calcium and phosphate
 - Biomimetic of natural salivary protein **statherin**
- CPP attracts calcium -> attracts phosphate -> attracts fluoride -> becomes a soluble nanocomplex of amorphous ions called CCP-ACP
 - Provides a pool of phosphate and calcium to maintain supersaturation of saliva
 - Also localizes ACP to the tooth surface
 - Can stabilize 100x more calcium and phosphate than is normally possible in solution
- Can also come as CCP-ACFP (with fluoride)



Mechanisms

- **Desensitization**

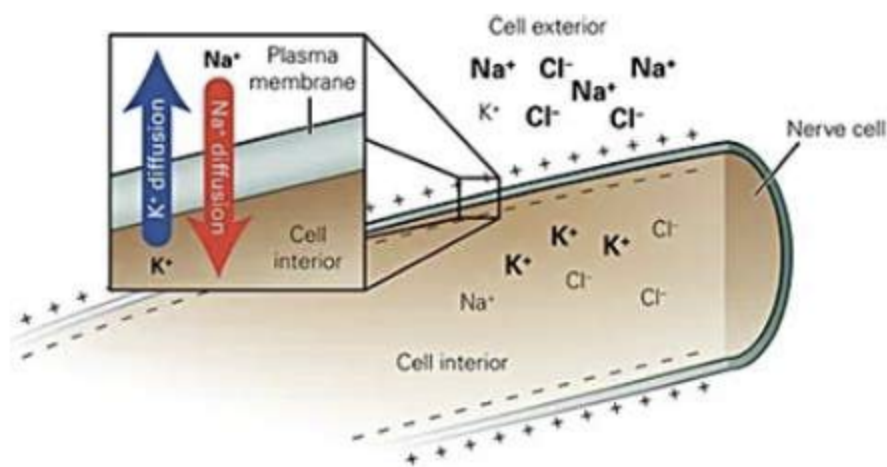
- **Promote formation of peritubular dentin, inside dentinal tubules, to partially block effect of external stimuli**
 - However in multiple studies, KNO₃ produced it's desensitizing effect FASTER than CCP-ACP

- **Bacteriostatic**

- CCP-ACP binds to biofilm complex, saturating it with calcium and phosphate ions
 - **Increases local pH in biofilm, promoting neutralizing effect**
 - Biofilm processes CCP into ammonia -> increases pH
 - **Masks the receptors on salivary molecules that allow bacterial adhesion**
 - Therefore slows down growth and CFUs of cariogenic bacteria
 - Measured effect in decreasing amounts of *Streptococcus mutans*
- Possibly helpful during active acid erosion due to pH effect

Potassium Nitrate (KNO₃)

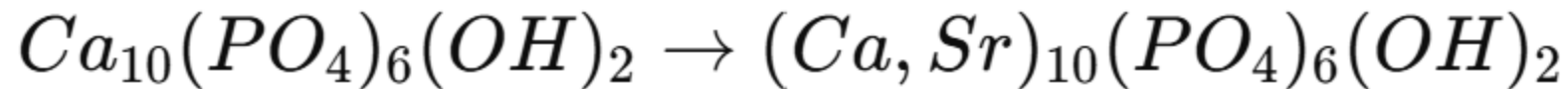
- **Depolarizes the nerves** within the dentinal tubules and inhibits their ability to transmit pain
- Most patients experiencing sensitivity have a reduction in their symptoms after a **2-week use** of potassium nitrate-containing dentifrice
 - Sensodyne
 - "rapid relief" has a gel scaffold that holds the KNO₃ and stannous fluoride in place better (more substantive)
 - Research shows application of FI varnish at 3 month intervals has same efficacy and speed of sensitivity relief



Strontium



- Reduction of dentinal hypersensitivity
- Works primarily by occluding dentinal tubules
 - Precipitates inside the tubules, forming insoluble (or minimally soluble) complexes that create a plug like barrier
- Strontium can also replace calcium in HA
 - Makes HA slightly less soluble, increasing acid resistance and making the tubule plug more stable
- Mineralized barrier takes repeated use to thicken over time and become effective
- But
 - Can be removed by vigorous brushing or acid etching
 - Less dramatic effect than some alternatives (SDF, glutaraldehyde + HEMA, bioactive glasses, etc)



Nanohydroxyapatite (nHA)

- 1970s: synthetic hydroxyapatite created by NASA to help restore enamel and bone loss of astronauts after gravity free missions
- 1978: private company Sangi buys the patent from NASA and develops it for commercial dental use
- 1980: commercial product launches in Japan
- 1993: Approved in Japan as an anti caries agent
- 2003: particle size decreased to 50nm, making it more effective at penetrating surface enamel

"Micro-HA particles are about 5–10 microns in size, and this is significantly larger than enamel HA and dentinal tubules."

Hydroxyapatite in Oral Care Products—A Review 2021 Aug PMID: 34500955

Nano

VS.

Micro

Mission: To advance the science and art of health through
accountability

These teaching materials are the property of California Northstate University and

Nanohydroxyapatite (nHA)

- Very similar in morphology, structure and crystallography to natural apatite crystals in enamel
- Nano size
 - increases surface area
 - Increases reactivity
- Calcium phosphate reservoir
 - Fluoride promotes remineralization by the building blocks present in saliva.
 - nHA IS the building block itself (does not need additional Ca and PO₄ for remineralization)
- Research shows equal (sometimes better) remineralization end result as Fluoride
 - But does not have better critical pH like FA
 - Does not remineralize FASTER like FA
- Combination of nHA and F⁻ increased the effectiveness of both the nHA and the F⁻
- Dose effect plateaued after 10% nHA concentration.
 - Therefore that is concentration in most products

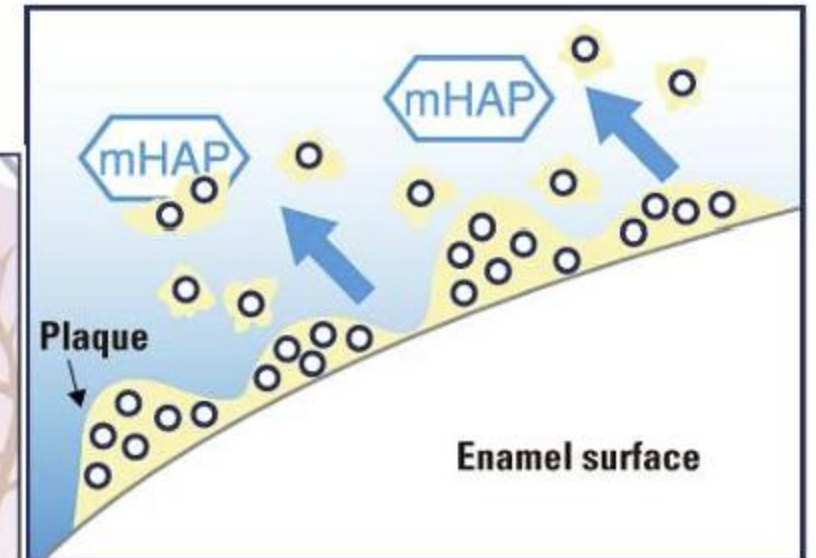
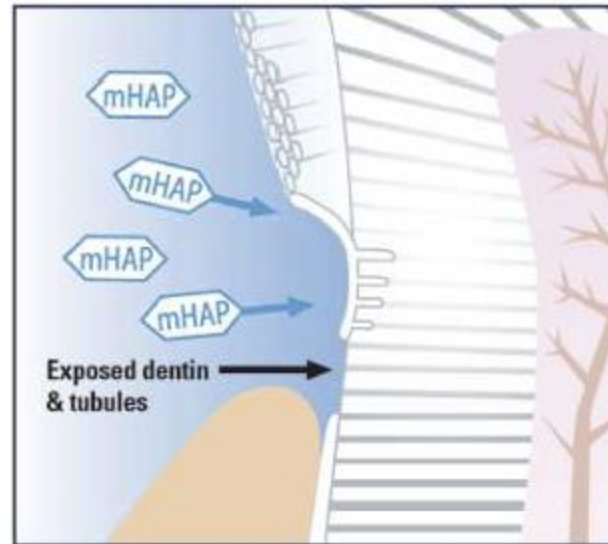
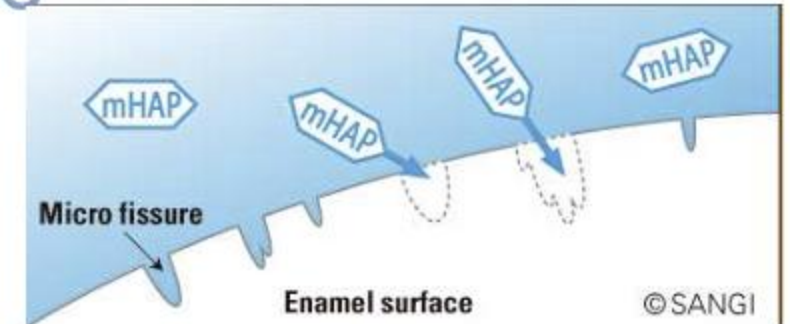
Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission

Nanohydroxyapatite (nHA)

- Mechanisms

- New layer of synthetic enamel at outer layer of tooth.
Directly fills micropores in the enamel surface!
- Protects against biofilm attachment and stains by
filling the micropores that promote them
- Absorbs plaque bacteria with a strong selectivity for
S. mutans
- Occlude dentinal tubules

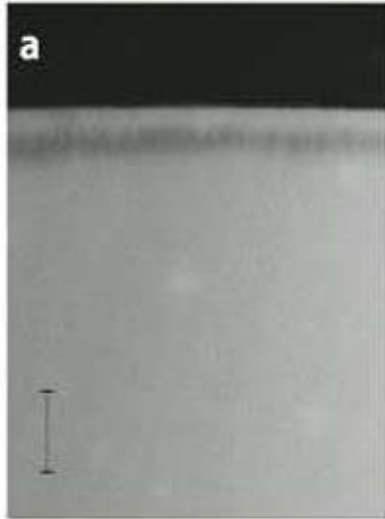


Mission: To advance the science and art of health through education, service, scholarship and social accountability

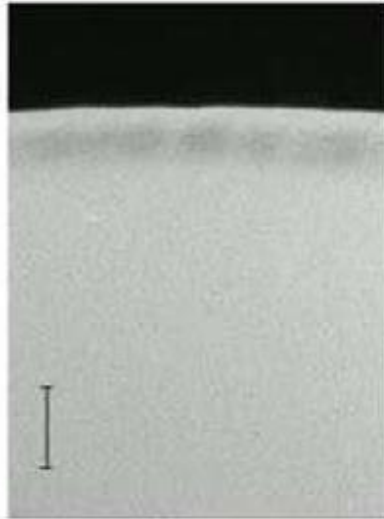
These teaching materials are the property of California Northstate University and may not be reproduced without permission

Nanohydroxyapatite (nHA)

***In situ* (clinical test) (Amaechi 2011, CRM)**



a. Demineralized enamel



b. Demineralized enamel after brushing with 1000ppm fluoride (NaF) toothpaste.



c. Demineralized enamel after brushing with nano<mHAP>® toothpaste.

Photograph (b) supplied by the author.

The Journal of Clinical Dentistry, 22(5), 139-143, 2011.

Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission

Nanohydroxyapatite

- Nanobiology is still an emerging fields. Changing scale can completely change the biochemical properties of a product
- EU has expressed concerns with:
 - Mucosal penetration
 - Nanoparticles may be small enough to enter systemic circulation through mucosa.
 - So far, studies suggest this is a very minimal risk
 - Inhalation
 - Nanoparticles may be breathed in and enter the lungs
 - Considered a more plausible concern than mucosal penetration
 - Is more of a risk with cosmetics. Aerosolization of toothpastes is not significant (but maybe with electric toothbrushes).
 - No significant clinical evidence of harm, but theoretical concern
 - Bioaccumulation
 - Expected to deposit in tissues and accumulate over time. No longer term studies available. Might be easily handled by the body in the same way as normal calciums and phosphates.
 - In EU, nanoparticles require special label on packages for consumer awareness



SLS- Sodium Lauryl Sulfate (SDS too)

- Most frequently used detergent in commercial dentifrices
 - Also in shampoos and other products.
- Extracted from coconut and palm oils
- Has anti-microbial properties and is responsible for the **foaming** action that gives a **perception** of cleaning
 - binds with bacterial protein and disrupts the bond between bacteria and the tooth surface
 - Adsorption to bacterial surface may interfere with cell wall integrity
 - subsequent leakage of cellular components.
- Is slightly substantive
 - Binds to calcium. Persists in places where calcium persists (oral cavity). However easily washed away by saliva.
 - Also explains its negative effects
 - **Irritating to skin and can cause contact dermatitis**
 - In sensitive persons, removes the protective mucin layer present in the soft tissues of the oral cavity. This compromises the integrity of the oral mucosa, resulting in ulceration of soft tissues

Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission

SDS/SLS Oral Sloughing

Typically asymptomatic incidental finding





Chewing gum (of any kind)

Pros

- Stimulate salivary flow
 - Body thinks its food that needs to be digested
 - Salivary flow increases oral pH
 - Presumes health/capable salivary glands
 - Other methods may be better at stimulating salivary flow
- Accelerate removal of carbohydrates that accumulate from food intake (because they stick to gum)
- Help with GERD symptoms

Cons

- **Considered a parafunctional habit**
 - Potential for TMD exacerbation
 - Potential for headaches
 - "Overuse and misuse"
- Potential for damaging temporary/ailing dental restorations
- If sweetened with fermentable carbohydrate, can be cariogenic

Chewing gum (of any kind)

- Recommended use
 - 15-20 minutes
 - Immediately after eating to help clear carbohydrates and continue salivary production for buffering effect
 - Is that how most people chew gum? After eating?
- Average patient is not recommended to add gum to their routine
 - But if you are in the habit of chewing gum ANYWAY...choose
 - Sugar free gum can substitute or add:
 - Xylitol
 - CPP-ACP
 - Fluoride

Xylitol- History

- 1890s: Discovered xylitol in Finland as an extract from Birch trees
- 1940s: WWII sugar rations motivate Finland to use xylitol more at home since sugar was going to the war front
- After WWII: Finland went back to using sugar. Noticed negative health changes from going back to sugar (increase in diabetes and tooth decay)
- 1960s: Began recommending that diabetics use xylitol to lower glycemic index
- 1963: FDA approves xylitol as sweetener and additive in USA
- 1970s: Studies finding that replacing sugar with 5-10g/day xylitol resulted in 40-80% less tooth decay and 50% less plaque buildup



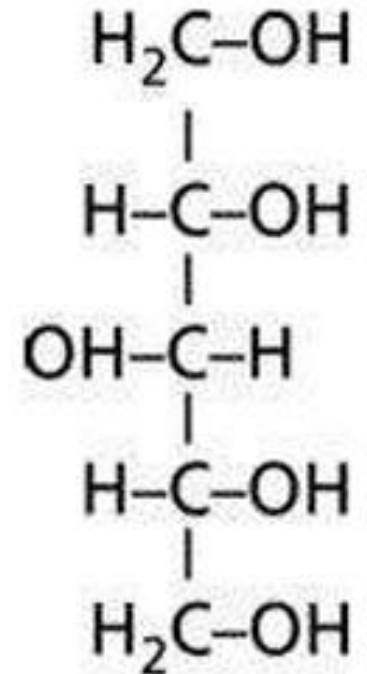


In 1967, the National Institute of Dental Research's National Dental Advisory Council (US) identified three main areas of emphasis to inform priorities to eradicate caries: Reducing the virulence of bacteria once exposed to sugars, fluoride delivery, and of most concern to the sugar industry and dietary modification.^[23] Similarly, xylitol has been in debate as a caries preventive strategy, and research has been carried for the past three decades. There is overwhelming evidence that sugar in the diet is the prime culprit in causing dental caries.^[24] Replacement of sugar with xylitol may be an effective strategy in preventing dental caries, but the results of this review do not indicate that prevention is worthwhile only with xylitol.

o and social

Xylitol - Mechanism

- Xylitol is a futile non fermentable carbohydrate that is bacteriostatic via metabolic sabotage
 - It is transported into microbes such as *S. mutans* but not metabolized any further
 - The cell then has to use energy to expel it back out
 - This futile use of ATP results in:
 - Reduced bacterial growth
 - Lower acid production
 - Metabolic stress
 - Also reduces EPS production which reduces enamel adhesion
- Clinically, the anticaries effect as a futile carbohydrate is modest
 - Most beneficial by substituting and reducing the amount/frequency of fermentable carbohydrate exposure.
 - Also very beneficial as a salivary stimulant



Xylitol - Concerns

- For a modest to moderate anticariogenic effect, recommended intake is 5-10grams/day over 3-5 exposures.
- However, xylitol is poorly absorbed. A significant portion of the administered dose can reach the colon, resulting in:
 - Fermentation by gut bacteria -> gas, bloating, and cramping
 - Osmotic effect -> pulls water into lumen -> diarrhea
- Some people notice these symptoms at 10-15g/day. Most severe GI stress occurs at 30+ grams/day.
- Emerging research suggests negative cardiovascular impact with high levels of xylitol in foods
 - High xylitol levels in plasma associated with increased cardiovascular events
 - Predominantly a concern in people who consume a lot of processed "sugar free" foods
 - Better to practice moderation than substitution
 - Recommended to keep total exposure under 10g/day.
 - Xylitol in oral products still considered safe due to low ingested dose.
- Not yet tested for safety during pregnancy or lactation
- Toxic to dogs, even in small doses

Other Common Sugar Substitutes

- Erythritol
 - Has some similarities to xylitol
 - Not metabolized by bacteria. But also not even taken in by bacteria (no futile metabolic effect)
 - Can inhibit biofilm adhesion
 - Is well absorbed in the small intestine. Unlike xylitol, does not have significant GI symptoms.
- Sorbitol
 - Very common in most labeled sugar free products
 - IS a fermentable carb and IS cariogenic. But very low cariogenic potential
 - Safer than traditional sugars (sucrose, glucose, etc) but does not have any anti caries effect. Just very low acid production.
- Stevia
 - Also a plant derived sweetener, like xylitol
 - Non cariogenic (non fermentable by oral bacteria)
 - But no active anti caries effect. Just non cariogenic, not anti cariogenic.
 - No GI fermentation
 - Taste/flavor can be a limitation
- Allulose
 - Can be found naturally but most of what is in food products is synthetically produced
 - Derived from fructose/corn-syrup.
 - Commonly used in Keto products
 - Non cariogenic but not anti cariogenic

Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission

A rinse is just a vehicle
of delivery.

What active ingredient
are you trying to
delivery to which site?

Is a rinse the best
vehicle of delivery for
that ingredient?

At what pH and timeline
does that ingredient
work effectively?



Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission

Why are some mouthwashes acidic?

- Stability in solution
- Facilitates binding of fluoride to enamel/remineralization
 - Fluoride is more bioavailable at slightly acidic pH
- Is acidic good or bad?
 - Does the patient have recession?
 - Are we concerned with enamel critical pH? Or dentin/cementum critical pH?
 - Are they using mouthwash as their primary source of fluoride or are they getting it elsewhere?
 - Are they using the mouthwash as a buffer? Or other reason?
 - Is it long term use or temporary?
 - After oral surgery
 - After periodontal treatment
 - Palliative for ulcers/mucositis

- Note that these are very different rinses with different active ingredients
 - Peridex is for biomass reduction and not fluoridated.
 - Commonly Rx for periodontal treatments
 - Biotene is for xerostomia (comfort), not anticaries
 - Therabreath is "oxyd-8" which is similar to sodium chlorite
- Looking at pH alone is insufficient
 - Specially given that different active ingredients may be more effective at different pH
 - However an extreme caries risk patient, for which pH control is a CAMBRA recommended treatment, should avoid an acidic mouth rinse in most cases

Chlorhexidine (CHX)

- **Chlorhexidine** is a **synthetic bisbiguanide compound**—it's not derived from a plant or natural source. It was developed in the 1950s by chemists.
- Antibacterial agent, often delivered as a mouthrinse in dentistry.
- Chlorhexidine is positively charged and **binds to the negatively charged groups on bacteria, on the mucosa, and in the biofilm.**
 - Substantivity (binding and slow release) is significant compared to some other rinses.
 - This inhibits adhesion of microbes to teeth.
- Bacteriocidal at high doses
 - Disrupts cell membranes
- Bacteriostatic at low doses

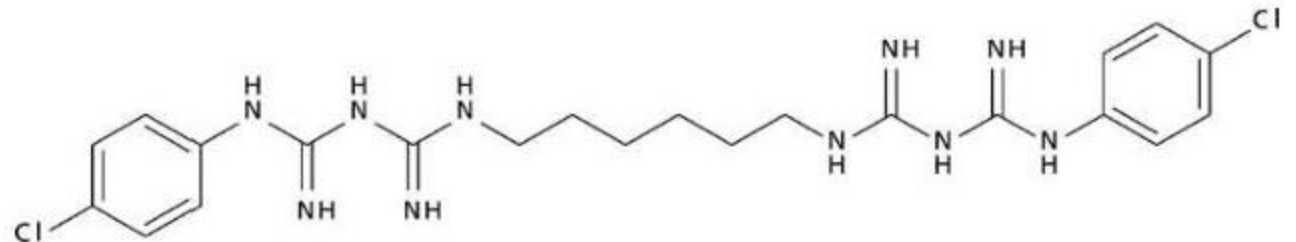


Figure 16.4 The molecular formula of chlorhexidine.

0.12% Chlorhexidine

AKA "Peridex" or "Periogard"

- Rx required
 - Light sensitive. Check expiration date and that kept in dark bottle.
- Indications
 - After 4-6 weeks:
 - Significant reduction in plaque/biomass
 - Moderate improvement in gingivitis
 - Effects can persist for a few months after discontinuation
 - May be used for biomass reduction in high risk/high plaque populations
- Contraindications
 - Commonly results in extrinsic staining after 2 weeks of use
 - Can be removed with polish/prophy
 - Concerns for antimicrobial resistance
 - Concerns for inhibition of fibroblastic activity
 - Altered taste and mucosal irritation with prolonged use (as with any antimicrobial)
- Alternatives
 - Alternative biomass reduction rinses are gaining popularity

Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission





Figure 1: Teeth stained by 0.12% chlorhexidine digluconate.



Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission

Chlorine Dioxide (ClO₂)

- NOT bleach...but can taste like pool water
- Is an oxidizer common used to disinfect water lines, surfaces, and instruments that is also available as a mouth rinse
- Strong antibacterial against prokaryotic cells (bacteria) but minimal effect on eukaryotic cells (host)
 - o Human fibroblasts are only affected at very high doses vs CHX
- Generates reactive oxygen species (ROS)
 - o Counteracts **halitosis** by direct ROS direct oxidation of volatile sulfur compounds (VSCs)
- But has **minimal substantivity compared to CHX**
- At oral rinse dosage: minimal staining, taste changes, or mucosal irritation

Chlorine Dioxide

- Closys
 - Stabilized chlorine dioxide
 - One bottle system
 - OTC
 - Fair taste
- Two bottle systems
 - Activated chlorine dioxide
 - gas
 - Slight pool water taste
 - Mix and use right away
 - Dental office only
 - Not Rx but sometimes not sold OTC



CHX vs ClO₂

Short- and long term antibacterial effects of a single rinse with different mouthwashes: A randomized clinical trial

Anna Herczegh^{a,*}, Boglárka Csák^a, Elek Dinya^b, Anna Moldován^c,
Ágoston Ghidán^d, Barnabás Palcsó^e, Zsolt M. Lohinai^a

- Objectives

- Examine the change in *S. mutans* and the total bacterial load in human saliva *in vivo* after a 1 rinse with diff. rinses.

- Material and methods

- One mL of unstimulated saliva was collected from volunteers with poor [oral hygiene](#) at **baseline** and **5 min** after a **1-min rinsing** with 0.0015% [chlorine dioxide](#); ClO₂, [Listerine Total Care](#)®, Corsodyl® (0.2% chlorhexidine-digluconate; **CHX**), or **BioGate Si*CLEAN** for bacterial investigation.
- In a second study, volunteers rinsed with **0.003% ClO₂ or CHX for 1 min**, and saliva was collected at **baseline**, after **5 min**, and after **90 min**.
 - After plating, the total plate and *S. mutans* colony numbers were determined.

- In the first study, **ClO₂ and CHX similarly reduced both total germ and *S. mutans* numbers**, while [Listerine Total Care](#)® decreased only the *S. mutans* counts. BioGate Si*Clean had no effect on either the total germ or *S. mutans* numbers.
- In the second study, an **increasing tendency toward bacterial [regrowth](#) was observed with CHX after 90 min compared to the 5-min value**, while no change was measured after ClO₂ rinsing



Biomass Reduction

Broad spectrum antimicrobial rinses (such as 0.12% Chlorhexidine or chlorine dioxide) are generally used for temporary biomass reduction in high risk individuals. The ecological plaque hypothesis model does not support broad and long term destruction of the oral microbiome.

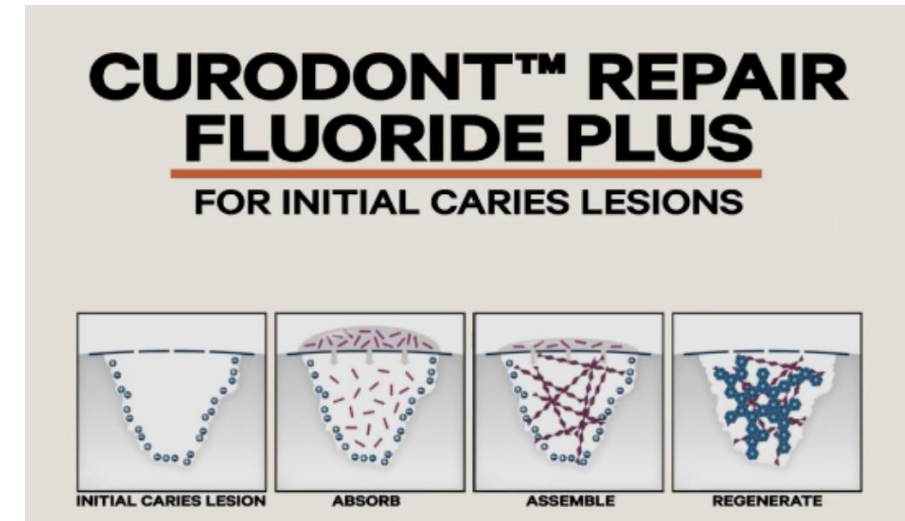
- Gingivitis control
 - To help with restorative outcomes (crown impression, composite isolation, etc)
 - But research does not support any meaningful improvement in periodontal diagnoses
- Short term post surgical use
 - Minimize exposure of open wounds to high bacterial oral environment
 - Generally used for 1-2 weeks
- High risk/low compliance situations
 - When mechanical biofilm disruption is less achievable such that plaque levels and natural dispersion is high.
 - Ex: Non compliant ortho patients, limited manual dexterity,

Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission

Self-Assembling Peptides

- Currently available products using P11-4: A short synthetic peptide designed to penetrate the depth of a non cavitated enamel lesions (ICDAS 1 and 2) and self assemble into a scaffold to support remineralization
- Goal is for remineralization to occur inside the lesion along the scaffolding (vs. Enamel surface with minimal penetration)
- Clinically applied by a professional
 - Clean surface and verify non cavitated and not in dentin
 - Etch or condition surface to open porosities for infiltration
 - Apply P11-4 solution, allow diffusion and self assembly
 - Recommended to combine with fluoride therapy, improved home hygiene, and dietary counseling
- New product and class of action. TBD



Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission

Emerging Research Categories

- Inoculation
 - If caries is an ecological dysbiosis and not a pathogen specific disease, is vaccination against specific pathogens the goal?
- Microbiome modulation
 - Pre/pro/post biotics
 - Shift or maintain ecology of the biofilm towards non carious microbes
 - Restore ecologic balance vs. Eradicating biofilm
 - Functional modulation
 - Influence the biofilm towards producing less acid, more base
- Virome and phage therapy
 - Bacteriophages to influence biofilm makeup, gene transfer, and virulence
- Precision antimicrobials
 - CRISPR based targeting to selectively delete or silence genes in cariogenic species
- Host modulation
 - Instead of targeting the biofilm, target other host factors
 - Salivary volume and composition, immune response, dietary drivers,
 - Personalized risk based prevention



Mission: To advance the science and art of health through education, service, scholarship and social accountability

These teaching materials are the property of California Northstate University and may not be reproduced without permission